

Mock Test 2

Read the following with your child:

1. This is a multiple choice paper, in which you have to mark your answer to each question on the separate answer sheet. You should mark only one answer for each question.
 2. Draw a firm line slowly through the rectangle next to your answer like this . If you **make a mistake**, rub it out as completely as you can and mark your new answer.
 3. Be sure to place your pencil in the correct section of the answer sheet. Mark your answer in the box that has the same number as the question in the booklet.
 4. You may find some of the questions difficult. If you cannot do a question, **do not waste time on it but go on to the next**. If you are not sure of an answer, choose the one you think is best.
 5. **Work as quickly and as carefully as you can.**
-

Familiarisation

1

How many 20p coins can I get in exchange for £4.60?

- A 12 coins B 22 coins C 23 coins D 46 coins E 92 coins
-

2

A bar chart tracks monthly rainfall. The vertical axis rises in increments of 15 mm. The tallest bar (November) reaches exactly 6 increments high.

How many millimetres of rain fell in November?

- A 60 mm B 75 mm C 80 mm D 90 mm E 105 mm
-

3

Rearrange all of the digits 4, 0, 9, 3, 7 to create the smallest possible odd number.

- A 03479 B 30479 C 30497 D 30749 E 97430

4

Subtract one hundred and nine from three thousand and two.

A 2893

B 2895

C 2903

D 2913

E 3111

5

Which of the following numbers is a factor of both 48 and 72, and is also a square number?

A 4

B 6

C 8

D 12

E 16

6

An analogue clock face shows the time as 08:20.

What is the size of the smaller angle formed between the hour hand and the minute hand?

A 120 degrees

B 130 degrees

C 140 degrees

D 145 degrees

E 150 degrees

7

A recipe for 6 people requires 450 g of flour and 120 g of sugar. Sanjay wants to adapt the recipe for 9 people.

How many grams of sugar will he need?

A 160 g

B 180 g

C 200 g

D 240 g

E 270 g

8

What is the value of the missing digit in this calculation?

$$\begin{array}{r} 5 \blacksquare 3 \\ + 289 \\ \hline = 862 \end{array}$$

A 5

B 6

C 7

D 8

E 9

9

Calculate: $(8 \times 7) - (36 \div 4) + 5^2$

A 38

B 42

C 62

D 72

E 76

10

The temperature inside a freezer is -18°C . The temperature rises by 7°C , then drops by 4°C once power is restored.

What is the final temperature inside the freezer?

A -29°C

B -21°C

C -15°C

D -11°C

E -7°C

11

A wooden plank measures 5.4 metres. A carpenter cuts off three identical pieces, each measuring 85 cm.

What is the length of the remaining piece of the plank?

A 2.55 m

B 2.85 m

C 3.15 m

D 4.55 m

E 4.85 m

12

A bucket contains 4 red balls, 5 blue balls, and 3 green balls. One ball is picked at random.

What is the probability that it is NOT blue?

A $5/12$

B $7/12$

C $1/4$

D $1/3$

E $2/3$

13

Which of the following is equivalent to $\frac{13}{40}$?

A 0.13

B 0.325

C 0.35

D 3.25

E 0.0325

14

■ – 17.4 = 34.85

What number must replace the box?

A 17.45

B 51.25

C 51.89

D 52.25

E 52.45

15

A delivery van sets off at 07:45. It takes 1 hour 55 minutes on its first route, stands idle for 25 minutes, then takes 1 hour 15 minutes for its final delivery.

At what time does the van finish its work?

A 11:20

B 11:00

C 10:45

D 11:40

E 12:00

16

A triangle has a horizontal base measuring 14 cm and a vertical height measuring 8 cm.

What is the area of the triangle?

A 22 cm²

B 44 cm²

C 56 cm²

D 112 cm²

E 120 cm²

17

A map has a scale of 1 : 50,000.

A road measures 6 cm on the map.

What is its actual physical length in kilometres?

A 3 km

B 30 km

C 0.3 km

D 300 km

E 1.2 km

18

Find the missing value:

$$\frac{4}{7} = \frac{24}{42} = \frac{\blacksquare}{63}$$

A 32

B 34

C 36

D 38

E 40

19

What is the value of x if: $5x + 13 = 3x + 27$

A 5

B 7

C 9

D 14

E 20

20

A sculpture is made of a cube (side 3 cm) joined to a cuboid (4 cm $\times$ 5 cm $\times$ 2 cm).

What is the total combined volume of the sculpture?

A 40 cm³

B 49 cm³

C 58 cm³

D 67 cm³

E 72 cm³

21

Hannah scores 17 out of 25 on a biology test.

What percentage did she achieve?

A 34%

B 58%

C 64%

D 68%

E 72%

22

A regular hexagon has a total perimeter of 42 cm.

What is the length of one individual side?

A 6 cm

B 7 cm

C 8 cm

D 9 cm

E 12 cm

23

Look at this sequence of numbers:

3, 7, 15, 31, 63, ...

Which rule accurately describes how to find the next term?

A Add 4 each time

B Multiply by 3 then subtract 2

D Square previous and subtract 2

C Multiply by 2 then add 1

E Add the two previous numbers

24

What is the total sum of all the prime numbers between 20 and 40?

A 83

B 112

C 120

D 131

E 142

25

Entry fees: Adult £14.50, Child £9.20.

A family of 2 adults and 3 children visits. They pay with a £100 note.

How much change do they receive?

A £43.40

B £44.60

C £55.40

D £56.60

E £62.40

26

A spinner has 8 equal sectors: 3 Red, 4 Blue, 1 Yellow.

The spinner is spun 160 times.

How many times would you expect it to land on Blue?

A 20 times

B 40 times

C 60 times

D 80 times

E 100 times

27

An isosceles triangle has one angle measuring 110° .

The remaining two angles are equal.

What is the size of one of these smaller angles?

A 35°

B 45°

C 55°

D 70°

E 80°

28

Which of the following numbers is a common multiple of both 6 and 9?

A 3

B 12

C 15

D 45

E 54

29

A storage box holds up to 36 books.

An online bookstore has 1,460 books to pack.

How many boxes will be needed to pack all the books?

A 40 boxes

B 41 boxes

C 42 boxes

D 43 boxes

E 44 boxes

30

Select the highest common factor (HCF) shared by 36, 60, and 84.

A 4

B 6

C 12

D 18

E 24

31

A pair of running shoes is reduced by 25% in a sale.
The sale price is now £45.

What was the original price of the shoes before the sale?

A £56.25

B £60.00

C £62.50

D £75.00

E £90.00

32

$0.045 \times 12 = \blacksquare$

A 0.54

B 5.4

C 0.495

D 4.95

E 0.054

33

A raffle gives winning tickets to multiples of 7 that are greater than 50 but less than 90.

Which of the following ticket numbers could be a winner?

A 49

B 67

C 77

D 81

E 91

34

Convert 4.35 litres into millilitres.

A 435 ml

B 4350 ml

C 43500 ml

D 0.435 ml

E 43.5 ml

35

Four items on a scale:

Item A = 0.65 kg, Item B = 320 g, Item C = 1.05 kg, Item D = 480 g

What is the total combined mass in kilograms?

A 2.1 kg

B 2.3 kg

C 2.5 kg

D 2.7 kg

E 3.1 kg

36

Find the median value of this data set:

12, 5, 19, 7, 14, 12, 8

A 7

B 8

C 11

D 12

E 14

37

A school bus drives 5 km directly North, then 12 km directly East.

What is the straight-line distance from its final location back to the start?

A 13 km

B 15 km

C 17 km

D 23 km

E 60 km

38

Three vertices of a square are at:

(2, 1), (6, 1), (2, 5)

What are the coordinates of the fourth missing vertex?

A (5, 6)

B (6, 5)

C (6, 6)

D (5, 5)

E (1, 5)

39

$3.7 \times 0.4 = \blacksquare$

A 1.48

B 14.8

C 0.148

D 1.38

E 13.8

40

A journey of 180 miles takes exactly 3 hours and 45 minutes.

What was the average speed in miles per hour?

A 45 mph

B 48 mph

C 50 mph

D 52 mph

E 55 mph

41

A rectangular lawn measures 15 m by 8 m. A garden path takes up 14 square metres.

What is the total area covered in grass turf?

A 96 m²

B 106 m²

C 120 m²

D 134 m²

E 144 m²

42

What is the decimal equivalent of 7/20?

A 0.72

B 0.35

C 0.45

D 0.175

E 0.035

43

A kite shape has two sides of length x and two sides of length y .

Which formula represents the total perimeter P ?

A $P = x + y$

B $P = xy$

C $P = 2x + 2y$

D $P = x^2 + y^2$

E $P = 2x \times 2y$

44

A Venn diagram: left circle = square numbers up to 150,
right circle = cube numbers up to 150.

Which number must be placed in the overlapping shaded section?

A 16

B 27

C 36

D 64

E 125

45

A garden plot measures 12 m by 8 m.
Paving tiles measure 0.5 m by 0.5 m.

How many paving tiles are needed to cover the entire garden?

A 96 tiles

B 192 tiles

C 240 tiles

D 384 tiles

E 400 tiles

46

Find the mean value of these three algebraic expressions:

$2x - 3$, $6x + 5$, $4x + 4$

A $12x + 6$

B $4x + 2$

C $4x + 3$

D $3x + 2$

E $6x + 3$

47

160 children walk to school. On a pie chart the walking section has an angle of 120 degrees.

How many total children took part in the survey?

A 320 children

B 400 children

C 480 children

D 540 children

E 600 children

48

A rectangular crate has a total surface area of 150 cm^2 .

Every dimension is exactly doubled.

What will the new surface area be?

A 300 cm^2

B 450 cm^2

C 600 cm^2

D 900 cm^2

E 1200 cm^2

49

A gamer's mean score after 5 levels is 140 points.
After the 6th level, the overall mean rises to 145 points.

How many points did they score on the 6th level?

- A** 150 points **B** 160 points **C** 170 points **D** 175 points **E** 180 points
-

50

The coordinates of 4 vertices of a quadrilateral are:
($a-5$, b) (a , b) ($a-5$, $b+3$) (a , $b+3$)

What specific geometric shape is formed?

- A** Square **B** Rectangle **C** Parallelogram **D** Trapezium **E** Rhombus